

VERSION 2.0 · SUPERSEDES V1.0

System Analysis & Design Document

Realigned to the confirmed domain model, rules, and workflows

The technical blueprint, corrected against the four design stages completed since v1.0 — with the data design and API sections deliberately marked provisional rather than fixed ahead of sequence.

Product	MentorMAMA — Digital Mentorship Toolkit for Safer Midwifery Clinical Placement
Prepared for	AfyaVentures 2026 Cohort / JHUB Africa Attachment Programme
Prepared by	Sinaps Technology — Bouric Okwaro Enos (Ric), Co-Founder & Team Lead
Document type	System Analysis & Design Document, following the JHUB Africa "Requirements to Reality" methodology
Supersedes	SADD v1.0 (July 2026) — see §11 for the change log and the Document Reconciliation Record for rationale
Source basis	Developer Concept Note v1.0, derived from Dr. Carolyn Kerubo Nyariki's research proposal, JKUAT School of Nursing
Version	2.0
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1. Introduction & Purpose

1.1 What changed between v1.0 and v2.0

SADD v1.0 was written before the domain model existed. It went straight from the Concept Note's requirements to classes, tables, and endpoints — and in doing so it made a set of structural decisions that the subsequent design stages overturned. The most consequential: it modelled the mentorship relationship as a **junction table** (PlacementAssignment) rather than as an aggregate, which quietly meant that reassigning a mentor would overwrite the record of who had mentored the student until then.

v2.0 realigns this document to the four stages completed since:

Stage	What it established
Stage 1 — Business Discovery	The ecosystem: two independent institutions meeting at exactly one shared object; the data-ownership table
Stage 2 — Domain Model	Placement as aggregate root; assignments as sub-entities; bounded contexts; domain events; the placement lifecycle
Stage 3 — Business Rules v1.1	19 policies, 18 invariants, 34 workflow rules, the transition tables, the authorization model, the error vocabulary
Stage 3.5 — Use Cases	All eight workflows with guard order, transaction boundaries, events, and exception paths

Where this document and those stages disagree, **those stages win** and this document is the defect. That is the authority order set out in the Document Reconciliation Record §1.1.

1.2 Two sections are deliberately provisional

§6 (Data Design) and §8.3 (API Design) are marked provisional and must not be built against. Stage 5 (Database Architecture) and Stage 6 (API Design) have not yet been produced. Fixing those sections now would repeat exactly the mistake v1.0 made — specifying a schema and an endpoint list ahead of the design that should determine them. They are retained as early sketches, clearly labelled, so nobody mistakes them for a contract.

1.3 Constraints & assumptions carried from the Concept Note

These are unchanged, and remain authoritative on scope:

- **No patient-identifiable data anywhere in the MVP.** Structurally enforced: no such field exists to write to (INV-18). Open-text fields carry the instruction at the point of entry (POL-015).
 - The system must work on common Android phones over weak (3G/4G) connections, with draft-saving and retry sync on the two forms mentors complete on the ward floor.
 - Pilot scope: one or two maternity units linked to JKUAT placement sites; 10–25 mentors; 30–80 students; 8–12 weeks.
 - AI features are Phase 2 only, restricted to mentorship and training, never patient-specific guidance.
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2. What the System Is

MentorMAMA is a **governance layer over a relationship that already exists**.

A university sends a student into a hospital ward it does not control, expecting a stranger employed by that hospital to supervise, teach, and vouch for that student's clinical competence. That relationship today runs on goodwill, paper checklists, and whoever is on shift.

The two real customers are the **University**, accountable for producing competent midwives, and the **Hospital**, accountable for what happens on its wards. They jointly own an outcome and share no system. Everything else in the product is instrumentation on top of that.

This framing has one dominant architectural consequence, and it is the reason the authorization model is the hardest part of this build: **neither institution has any authority inside the other**. A Nurse Manager does not outrank a University Coordinator. They are non-overlapping jurisdictions meeting at one object — the Placement. Access control is therefore not a role hierarchy; it is a scope intersection.

3. Actor & Stakeholder Analysis

3.1 MVP actors – five human, one system

Corrected from v1.0, which listed seven actors including a CPD/Content Reviewer with MVP permissions. That role touches no part of the placement workflow, and the Concept Note itself defers regulator integration to Phase 2 (§19). It remains in the vocabulary; it is out of the authorization model.

Actor	Belongs to	Core responsibilities	Authorization boundary
Student	University (belongs to, not employed)	Confirm placement, complete orientation, receive mentorship, submit feedback and confidence self-assessments, raise concerns	Own placement records only. Never manages anyone
Clinical Mentor	Hospital (midwife, senior nurse, clinical instructor)	Complete training, complete induction per student, log mentorship sessions, review follow-ups	Placements where they are the active Mentor Assignment; own training record
Nurse Manager / Ward In-Charge	Hospital (operational manager)	Verify students and mentors, assign and reassign mentors, monitor induction and session activity, review and close escalations, pause and close placements, export facility reports	Facility-scoped, ward as a filter (AUTH-006)
University Coordinator	University	Create cohorts, enrol students, allocate students to facilities and wards, issue invitations, monitor placement quality, export institution reports	Institution-scoped; session metadata only, never free text ; never hospital staff records
Program Administrator	Platform / Programme (JHUB Africa)	Configure institutions, facilities, wards, users, modules and content; programme-wide analytics	Programme-scoped. Configuration authority is not domain authority — cannot violate an invariant

Actor	Belongs to	Core responsibilities	Authorization boundary
Notification Service	System actor	Dispatch email (now), SMS/WhatsApp (Phase 2) on domain events	Triggered by events only. Executes; never decides

Phase 2: CPD/Content Reviewer — reviews training content and confirms completion for CPD purposes.

3.2 Out-of-scope actors

Patients and guardians are not actors and have no interface; no patient-identifiable data is captured. Finance and billing actors are out of scope, as payments are excluded from the MVP.

4. Use Case Analysis

Superseded by Stage 3.5. The eight core workflows are specified there with guard order, transaction boundaries, domain events, and exception paths. This section retains only the catalogue and its traceability; it does not restate the flows, because a second, shorter version of a specification is a second source of truth.

ID	Use case	Primary actor(s)	Stage 3.5 workflow
UC-01	Invite and activate account	Coordinator / Admin (invite); all roles (accept)	W-01
UC-02	Set up institution, facility, ward, cohort	Program Admin, Coordinator	W-01, W-02
UC-03	Create placement and allocate ward	Coordinator	W-02
UC-04	Assign / reassign mentor	Nurse Manager	W-03
UC-05	Complete mentor training module	Clinical Mentor	W-03 precondition
UC-06	Complete labour ward induction and activate placement	Mentor, Nurse Manager	W-04
UC-07	Log a mentorship session (incl. offline retry)	Clinical Mentor	W-05
UC-08	Submit feedback, baseline and final assessment	Student	W-06
UC-09	Raise an escalation	Student, Mentor	W-07
UC-10	Review, resolve and close an escalation	Nurse Manager, Coordinator	W-07
UC-11	Complete placement, report and archive	Nurse Manager, system	W-08
UC-12	View dashboard and export report	NM, Coordinator, Admin	W-08
UC-13	Receive automated notification	All (via Notification Service)	Cross-cutting

5. Domain Model

5.1 Design approach – two corrections to v1.0

Correction 1: composition over role-inheritance is retained. A single `User` with a role attribute and role-scoped profile extensions, rather than a subclass hierarchy. This held up and is unchanged.

Correction 2: Placement is an aggregate, not a junction. v1.0's `PlacementAssignment` joined `Student` ↔ `Mentor` ↔ `Ward` ↔ `Cohort` in one row. Stage 2 replaced this with a `Placement` aggregate root containing **assignment sub-entities**, each with its own validity period:

```
Placement (aggregate root)
├─ Student Assignment      (exactly one active – INV-01)
├─ Mentor Assignment      (exactly one active, history preserved – POL-005, INV-12)
├─ Ward Assignment        (exactly one active, history preserved – POL-004)
├─ Induction Checklist     (one per placement, survives mentor reassignment)
├─ Mentorship Sessions    (belong to the placement, not to a person – INV-03)
├─ Pause Intervals        (first-class records, not a boolean)
└─ Progress               (computed, never stored as authored truth)
```

The reason this matters in one sentence: **assignments change independently, and the history is the product.** A mentor goes on leave in week 3; a student rotates ward in week 4. A junction row overwrites both facts, and the university loses the ability to say who was responsible for what.

Adjacent aggregates, deliberately **not** inside `Placement`:

Aggregate	Why it is separate
Institution (→ Cohorts) / Facility (→ Wards)	Organisational scoping; outlive any placement
Training Module (→ Lessons, Quiz, Completions)	Mentor training is facility-independent and outlives any single student relationship (POL-010)
Escalation (→ Reviews, Actions)	References a placement but has its own lifecycle and its own restricted visibility (INV-13)
Assessment (feedback, confidence, evaluations)	Evolves independently of the placement workflow; new survey instruments must not require placement changes

5.2 Bounded contexts

Context	Owns	Notes
Identity	User, credentials, scope resolution	One organisation per user (AUTH-003)
Institution	Institution, Cohort, student enrolment	University side
Facility	Facility, Ward, capacity	Hospital side
Placement (core)	Placement, assignments, pauses, state machine	The core domain
Induction	Checklist templates (versioned), checklists, items	Placement-bound
Mentorship	Mentorship Sessions, follow-ups	Placement-bound
Learning	Training Modules, quizzes, completions	Facility-independent

Context	Owns	Notes
Assessment	Feedback, confidence, evaluations	Immutable submissions
Safeguarding	Escalations, reviews, actions	Strictest access control
Analytics	Projections, dashboards, exports	Read-only, event-fed
Notification	Delivery only	Never decides (WFR-023)

5.3 Placement lifecycle

Ten states, seventeen transitions, defined authoritatively in Stage 3 §5.2. Summary: `Draft` → `Scheduled` → `Awaiting Student` → `Orientation` → `Induction` → `Active` → `Completed` → `Archived`, with `Active` ⇄ `Paused` lateral and `Withdrawn` reachable from every non-terminal state.

Two properties of this model that v1.0 lacked entirely:

- **A placement cannot become `Active` until induction is complete** (INV-02). This is the core product promise — no unoriented student practising on a labour ward — and it is enforced in a transaction, not by screen order.
- **Escalation has an independent lifecycle** (`Open` → `Under Review` → `Action Required` → `Resolved` → `Closed`) and never redefines a placement's state. A placement stays `Active` while a concern is investigated, and multiple escalations can coexist.

5.4 Value objects

Person Name · Phone Number · Email · Placement Period · Competency Score · Ward Capacity · Progress Summary.

`Progress Summary` is computed on read, never persisted as an editable field (WFR-022) — a correction to v1.0's `DashboardMetric` entity with stored `numerator` / `denominator` columns.

6. Data Design — PROVISIONAL

This section is not authoritative and must not be built against. The relational schema is the **Stage 5** deliverable and does not yet exist. What follows states only the constraints that Stage 5 must satisfy, which are already known from Stage 3. The ERD, data dictionary, and indexing strategy in SADD v1.0 §5 are withdrawn: they described the junction-table model that Stage 2 replaced.

Constraints Stage 5 must honour, each traceable to a rule:

Requirement	Source
Partial unique indexes guaranteeing exactly one <i>active</i> Student / Mentor / Ward Assignment per placement	INV-01, INV-12
Check constraints on all score ranges (0–100, 1–5)	INV-15
Immutability of submitted feedback and assessments, and of terminal-state placements, enforced in the database	INV-09, INV-14

Requirement	Source
No delete paths on operational tables; soft-archive only	INV-10
Denormalised <code>institution_id</code> / <code>facility_id</code> on Placement for row-level scoping, with an invariant guaranteeing they equal derived values	POL-006, INV-17
Ward capacity enforced under a row lock, not a read-then-write	INV-16, WFR-029
Append-only audit table carrying actor, role, scope, action, target, before/after — and never restricted field values	WFR-030, WFR-034
A transactional outbox table, written in the same transaction as the domain rows	WFR-026
Versioned induction checklist templates, so live checklists never mutate underneath a mentor	Stage 3.5 §6.3
Anonymous feedback with no placement or student reference	POL-014
Optimistic version column or row-lock discipline on Placement for all state transitions	WFR-028
Third normal form, with checklist items and scores as rows rather than JSON blobs — the data is audit and CPD evidence	Carried from v1.0 §5.1

One decision blocks Stage 5: **OD-01** — whether a pause extends the placement end date determines whether placement dates are mutable, and therefore whether planned and actual dates are separate columns.

7. System Architecture

7.1 Pattern

A 3-tier layered architecture — Presentation, Application, Data — right-sized for a 3–6 developer team and affordable managed hosting.

Tier	Responsibility	Stack
Presentation	Mobile-first PWA: dashboards, forms, checklists, local draft store and client outbox	React + TypeScript + Vite; Tailwind CSS
Application	Authentication, scope resolution, domain services, invariant enforcement, transactional outbox	Django + DRF
Data	Relational storage, projections, caching, file storage	PostgreSQL; Redis; S3-compatible object storage

7.2 Module architecture — deferred to Stage 4, with one conclusion already fixed

Stage 4 will define module boundaries from the command and event surfaces in Stage 3.5 Appendices A and B. One finding is recorded here because it affects the existing codebase:

The current backend app layout cannot express the domain. `backend/apps/` contains `sessions`, `induction`, `feedback`, `escalations`, `training`, `facilities`, `dashboards`, and `content_library` — but **no placements app at all**. The aggregate root that coordinates all eight workflows has no home, while three of its children each have their own. Additionally, `apps/sessions` collides in name with `django.contrib.sessions`. Stage 4 should

restructure around the bounded contexts in §5.2 and treat the existing app list as scaffolding to be replaced — cheap now, expensive once migrations exist.

7.3 Cross-cutting concerns

Concern	Approach	Rule
Scope resolution	Derived server-side from the authenticated principal on every request. Never read from a request body, and never trusted from a token claim — a claim is a cached copy that goes stale the moment a mentor is reassigned	AUTH-001
Row-level scoping	Applied at the data-access layer, not the view layer. An endpoint that forgets its filter returns nothing, not everything	AUTH-002
Field-level restrictions	Applied in serialisation, so no code path can leak a restricted field by returning a "full" object	AUTH-005, INV-13, POL-016
Offline support	Local drafts plus a client outbox with a <code>command_id</code> generated at form-open time; server-side idempotency; late replays validated against current state and surfaced to the user if they no longer apply	Stage 3.5 §2.4, §7.4
Side effects	Transactional outbox: events written in-transaction, dispatched after commit with retry. No external I/O inside a transaction	WFR-026, WFR-027
Notifications	Email in MVP; SMS/WhatsApp Phase 2. Payloads carry identifiers and a template key — never escalation content, individual feedback, or session free text	WFR-025
Exports	First-class, scope-filtered, field-restricted, and audited with actor, scope, filters, and row count	TX-09, WFR-032
Audit	Append-only on every write; reads audited for escalation content, individual feedback, and exports	WFR-030, WFR-032

7.4 Authentication & authorisation

Invite-only accounts — no self-service sign-up, since every account must be tied to a verified institution or facility. Authentication is email/phone plus password with invite redemption.

Corrected from v1.0: tokens carry **identity**, not authority. v1.0 put `facility_id / institution_id` claims in the JWT and checked them at middleware; that grants a reassigned or deactivated user their old access until the token expires. Scope is resolved per request from the user record.

Cross-scope access returns **404, not 403** (AUTH-004): a 403 confirms the record exists, which is itself a disclosure when the record is a placement at another facility.

The full action matrix is Stage 3 §9.2 and is not duplicated here.

7.5 Non-functional requirements

Carried forward from v1.0 unchanged — these held up.

Category	Requirement
Performance	Page load under 2s on 4G; facility-scoped dashboard queries under 1.5s; session-log form completable in under 3 minutes on a mid-range Android device
Availability	99.5% during pilot; daily backups; RTO under 4 hours

Category	Requirement
Security	TLS 1.3 in transit, AES-256 at rest; audit logs on writes, escalation reads, and exports
Accessibility	Responsive from 375px; 4.5:1 minimum contrast; full keyboard navigability
Privacy	No patient-identifiable data in the schema; anonymised or aggregate reporting by default
Scalability	Additional facilities and institutions onboard without structural change

8. Interfaces

8.1 Information architecture

Role-scoped zones, matching the authorization model.

Zone	Key screens
Public	Landing, login, invite redemption
Student	Dashboard, My Placement, Sessions Received, Feedback & Confidence, Raise a Concern, Profile
Mentor	Dashboard, My Students, Training, Induction Checklist, Session Log, Follow-ups, Unsynced Logs (new — OD-09)
Nurse Manager	Dashboard, Mentor Assignment, Placements (incl. pause/complete), Escalations, Facility Reports
Coordinator	Dashboard, Cohorts & Students, Placements, Institution Reports
Program Admin	Dashboard, Institution/Facility/Ward setup, User Management, Content Library, Programme Reports

8.2 UI principles

Mobile-first at 375px · progressive disclosure, since the session log and induction checklist are the time-critical forms · **explicit state design**: every screen defines empty, loading, error, offline-queued, and success states · accessibility as specified in §7.5.

The **offline-queued** state is new in v2.0 and is not optional: Stage 3.5 §7.4 established that a mentor's queued work can fail validation on late arrival, and the mentor must be able to see and resolve that rather than discover silently that a session was never recorded.

8.3 API design — PROVISIONAL

Not authoritative. Transport design is the **Stage 6** deliverable. SADD v1.0 §6.3's endpoint list is withdrawn — most visibly `POST /api/placements` for "create/reassign", which conflated two operations with entirely different guard sets and actors.

Stage 6 will derive the transport surface from the **command surface** in Stage 3.5 Appendix A. Two requirements are already fixed: every mutating operation accepts a client-generated idempotency key, and every error response uses the shared domain error vocabulary in Stage 3 §12 rather than a generic HTTP status.

9. MVP Backlog & Acceptance Criteria

9.1 Epics

Epic	Key story	Acceptance criterion
Identity & scope	Admin invites users; users log in	Permissions differ by role; cross-scope API access with a valid token returns 404 ; deactivated users cannot access the system
Institution & facility setup	Admin/Coordinator create institutions, facilities, wards, cohorts	Students and mentors correctly linked; ward capacity recorded
Placement lifecycle	Coordinator creates placements; NM assigns mentors	Placement moves only along the Stage 3 transition table; illegal transitions rejected with <code>InvalidTransition</code>
Mentor training	Mentor completes modules and quizzes	Completion, score, date, and failed attempts all recorded
Induction & activation	Mentor completes checklist	Placement cannot become <code>Active</code> with required items outstanding; activation and completion commit atomically
Session logging	Mentor records an interaction	Saves in under 3 minutes; rejected outside <code>Active</code> ; duplicate submissions idempotent; offline queue survives app restart
Feedback & assessment	Student submits feedback and confidence	Immutable once submitted; anonymous submissions carry no placement or student reference; mentor aggregates withheld below $n = 3$
Safeguarding	Student/Mentor raise a concern; reviewer resolves it	Five-state lifecycle enforced; <code>description</code> unreadable outside reviewer/submitter/Admin; unauthorised access returns 404 and is audited; reviewer $\neq$ subject
Completion & reporting	NM completes placement; Coordinator exports	Completion blocked without final assessment or with an open escalation; exports scope-filtered, field-restricted, and audited

9.2 System-level acceptance criteria

Carried from v1.0, plus four invariant-enforcement criteria that v1.0 could not state because the invariants did not exist yet:

1. Each of the five human roles completes their core workflow without developer support.
2. A mentor completes a module, passes a quiz, and sees completion status.
3. A mentor completes an induction checklist and logs a session in under three minutes on a mid-range Android phone.
4. A student submits feedback and a confidence rating from a phone.
5. A Nurse Manager sees incomplete inductions, inactive mentors, and unresolved escalations.
6. A Coordinator exports CSV/XLSX scoped to their institution.
7. The system runs on common Android phones and modern browsers, and retains drafts across connectivity loss and app restart.

8. Row-level and field-level access control are verified by test, including direct API attempts from another facility's valid token.
9. A session cannot be persisted for a non- Active placement, outside the placement window, or inside a pause interval — verified by test, not by UI inspection.
10. A placement cannot be activated with an incomplete induction, nor completed with a missing final assessment or an open escalation — verified by test.
11. Every write produces an audit entry, and every override records a reason — verified by test.
12. No workflow requires patient-identifiable data.

10. Risks & Open Decisions

10.1 Risks

Risk	Likelihood / impact	Mitigation
Weak ward connectivity disrupts session logging	High / medium	Local drafts, client outbox, idempotency keys, sub-3-minute form, visible unsynced queue
Late-arriving offline logs fail validation and are lost	Medium / high	Retained client-side and surfaced for human resolution; never silently discarded (OD-09)
Mentor engagement drops after novelty	Medium / high	CPD-eligible completion summaries; manager visibility into activity
Escalation content exposure	Low / severe	Field-level restriction in serialisation, reviewer conflict rules, audited reads, no content in any event or notification payload
Patient identifiers typed into free text	Medium / severe	Point-of-entry instruction and reviewer redaction. Residual, not solved — must be named in the data-governance sign-off
Invariants enforced only at the service layer	Medium / high	Safeguarding and integrity rules pushed to DB / TXN ; walking-skeleton spike before feature build to prove enforceability
Scope creep toward EMR features	Medium / high	MVP excludes patient data and clinical decision support (§1.3)
Schema built ahead of Stage 5	Medium / high	§6 and §8.3 marked provisional; no migrations until Stage 5 lands

10.2 Open decisions

Eleven decisions are tracked in the consolidated register (Document Reconciliation Record §7). Three affect immediate work: **OD-03** and **OD-06** block pilot data collection because they gate consent wording; **OD-01** blocks Stage 5 because it determines whether placement dates are mutable.

11. Change Log – v1.0 → v2.0

#	Change	Reason
1	Placement modelled as an aggregate with assignment sub-entities, replacing the <code>PlacementAssignment</code> junction	A junction row overwrites relationships on change; the history of who mentored whom, and when, is the product
2	Sessions, induction checklists, and progress reparented from <code>StudentProfile</code> / <code>MentorProfile</code> to <code>Placement</code>	A student has many placements over time; person-owned records cannot be told apart
3	Placement dates and <code>assigned_mentor</code> removed from <code>StudentProfile</code>	Same reason; a second placement would overwrite the first
4	Actor set reduced from seven to six; CPD Reviewer moved to Phase 2	That role touches no placement workflow, and the Concept Note already defers regulator integration
5	Placement lifecycle added (ten states, seventeen transitions)	v1.0 had no lifecycle; "active" would have become a boolean someone flips
6	Escalation given a formal five-state lifecycle, conflict-of-interest rule, and field-level access control	v1.0 described the safeguarding workflow narratively — the most consequential path had the least specification
7	<code>DashboardMetric</code> entity replaced by computed projections	Stored metrics drift from the facts they summarise
8	Transactional outbox introduced; notifications and projections moved out of the write path	An email timeout must not roll back a valid mentor assignment, and a sent email cannot be un-sent
9	JWT scope claims replaced by per-request server-side scope resolution	A token claim grants a reassigned user their old facility access until expiry
10	Cross-scope denial changed from 403 to 404	A 403 confirms the record exists
11	Field-level restrictions added: coordinator sees session metadata only; mentors see feedback aggregates at $n \geq 3$; escalation content restricted to reviewer, submitter, and Admin	Enforces the Stage 1 data-ownership boundaries, which v1.0 stated but did not implement
12	Ward capacity enforced as an invariant under a row lock	<code>max_students</code> existed in v1.0's data dictionary and was enforced by nothing
13	§5 Data Design and §6.3 API Design withdrawn and marked provisional	They preceded the domain model; replacing them now would repeat the error out of sequence
14	§3 use cases and §8 workflows superseded by Stage 3.5	A shorter second copy of a specification becomes a competing source of truth
15	Offline behaviour specified end to end, including idempotency keys, late-replay validation, and a mentor-visible unsynced queue	v1.0 said "draft-saving and retry sync", which understates a path that guarantees replays
16	Acceptance criteria extended with four invariant-enforcement tests	Rules that no test exercises are documentation, not constraints

#	Change	Reason
17	Backend app-layout finding recorded (§7.2)	The aggregate root has no home in the current scaffold, while three of its children do

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